# OpenTab Global

Family support, spendable at the corner shop, approved on a phone with no data, settled on Stellar.

Blockchain Innovation Challenge
powered by Stellar
Problem Space 3: Transnational Households

## THE PROBLEM

Money sent home often has to become cash at a pickup counter before it can buy groceries or medicine. Many recipients have a phone but unreliable or costly mobile data. Senders cannot easily earmark support for essentials, and neighbourhood shops carry informal credit risk.

## OUR SOLUTION

A remittance becomes a **shared family spending balance** held by a Soroban smart contract. The family member pays at participating local shops by approving each purchase on their phone, **without internet**. The online shop submits the approval; the contract pays the shop in one atomic transaction.

## HOW A PURCHASE WORKS

SENDER · ONLINE
**Funds**
the family balance
→
SHOP · ONLINE
**Invoice QR**
local price + exact charge
→
FAMILY PHONE · OFFLINE
**Signs**
that exact purchase
→
SHOP · ONLINE
**Submits**
the signed approval
→
STELLAR
**Verifies + pays**
once; replays refused

## OFFLINE AUTHORIZATION

The phone holds an Ed25519 key registered on chain by the sender. Approving signs a fixed 340-byte message binding the Testnet network, the contract, the token, family, shop, terminal, a random invoice ID, the exact amount, the quoted local price and an expiry. The contract rebuilds those bytes and verifies the signature: change any field and it fails.

## WHY STELLAR

- Soroban enforces the rules on chain: one-time invoices, per-family balances, key revocation, merchant binding.
- Stellar Asset Contracts let the contract hold and pay a dollar-denominated token.
- Low fees and ~5 s finality make per-purchase settlement practical.
- Native Ed25519 verification: the family member needs no Stellar account and no XLM.

## VERIFIED PROTOTYPE (STELLAR TESTNET)

- FamilyVault deployed: `CAUCH3MNJGIYDLOY3OB3YCKRCBWWY3J3HZB5QCHPJXUWXBMK5IABYLGW`
- Real signed payments at two shops from one family balance; e.g. tx `0b8d72ed…bb05f`
- Same approval replayed: refused by the contract. Overspend: refused.
- 34 contract tests · 13 protocol tests · 18 live API checks · 18 security checks · browser tests including signing in an emulated offline browser.

## WHAT MAKES IT DIFFERENT

- **vs. remittance apps:** spendable at the shop with no cash pickup.
- **vs. gift cards:** one balance across independent shops; rules enforced by a public contract.
- **vs. crypto wallets:** the shopper needs no data, no XLM, no crypto knowledge.

## ROADMAP

Physical-device testing → both-offline mode with funds reserved for one named shop (in development) → real USDC → licensed local payout partners → audit and pilots.

**Honest limits:** Stellar Testnet prototype. Settlement token is OpenTab Demo USD (OTUSD): no value, not USDC. Exchange rates are illustrative. No bank payouts. Unaudited. Built with AI assistance.